



University of Colorado **Boulder**

September 22, 2026

ASEN 5335: Aerospace Environment

Module 3: The Ionosphere

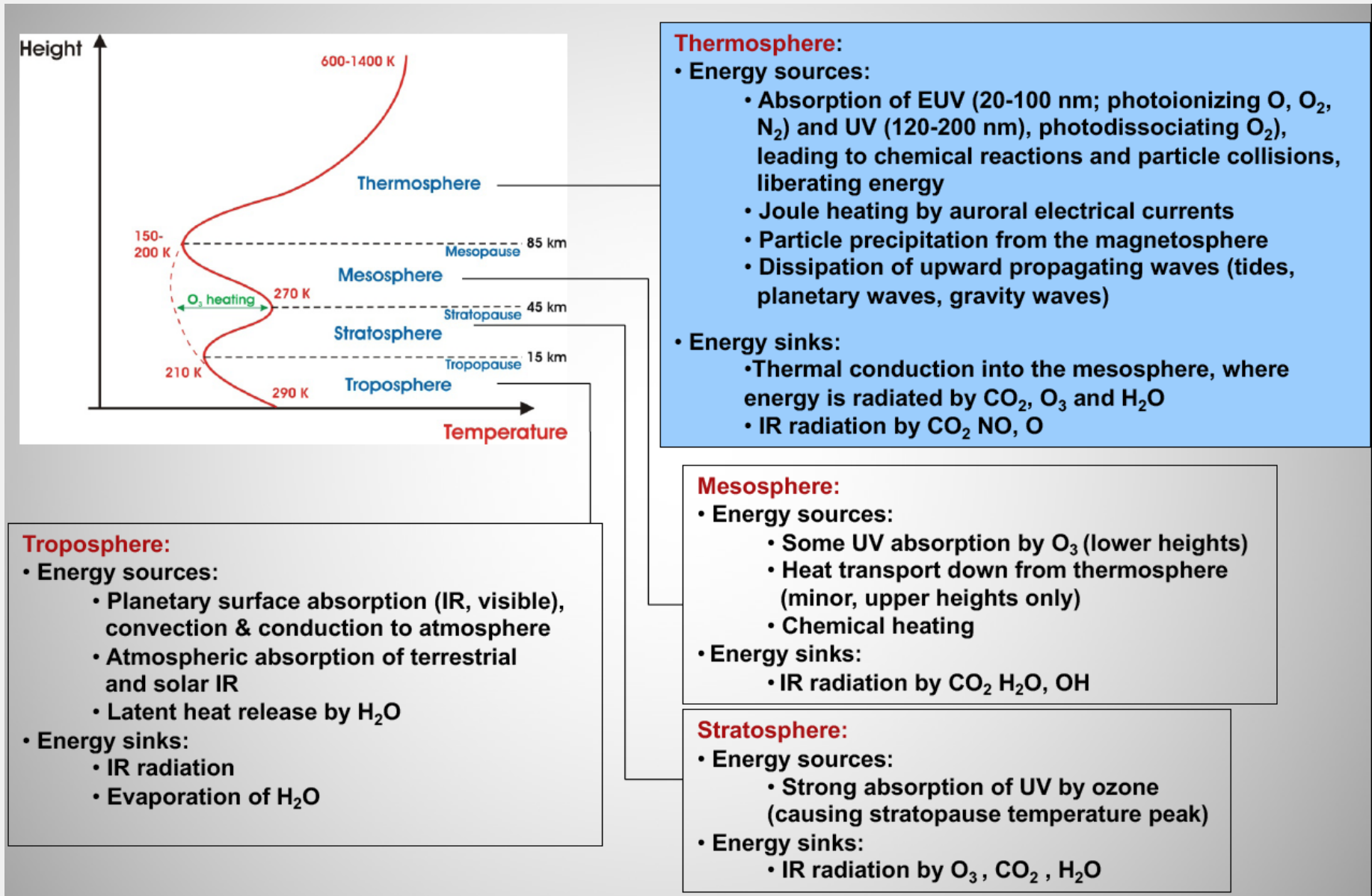
Xinlin Li

Goals of This Module

- ❖ **Understand basic physics of the Earth's ionosphere**
 - ❖ Origin, composition, layers
 - ❖ Variations: diurnal, seasonal, solar cycle, plus other anomalies
 - ❖ Basics of the plasmasphere
- ❖ **Understand effects of the Ionosphere on Spacecraft and technology**
 - ❖ Spacecraft charging
 - ❖ Radio communications and GPS
- ❖ **Learn about Ionospheric space missions, including ICON and AMPERE**

Origin of the Ionosphere

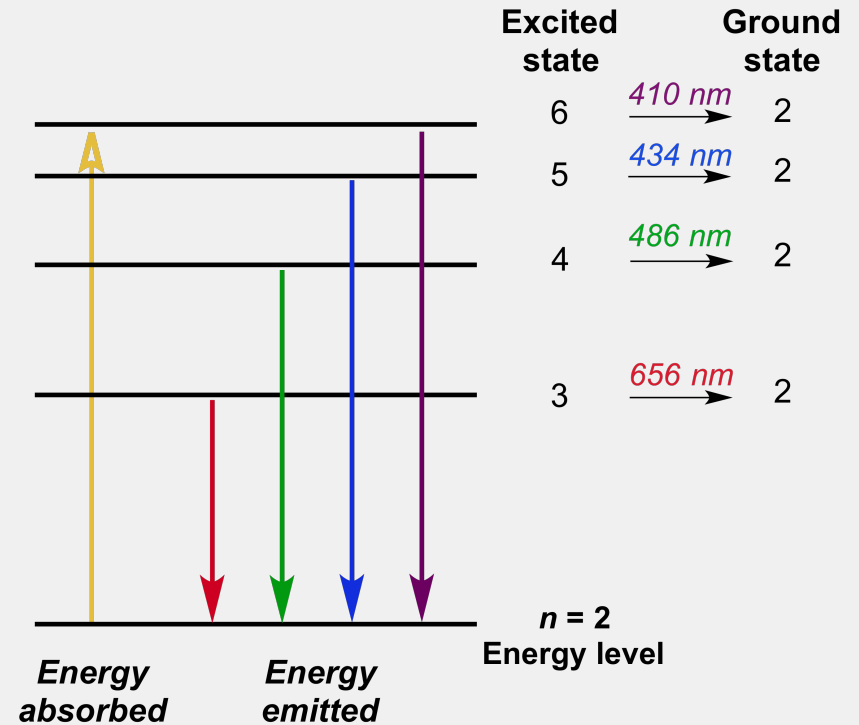
- ❖ The Ionosphere is a product of two regions we've covered already: the **Sun**, and the **Atmosphere**



Absorption in Earth's Atmosphere

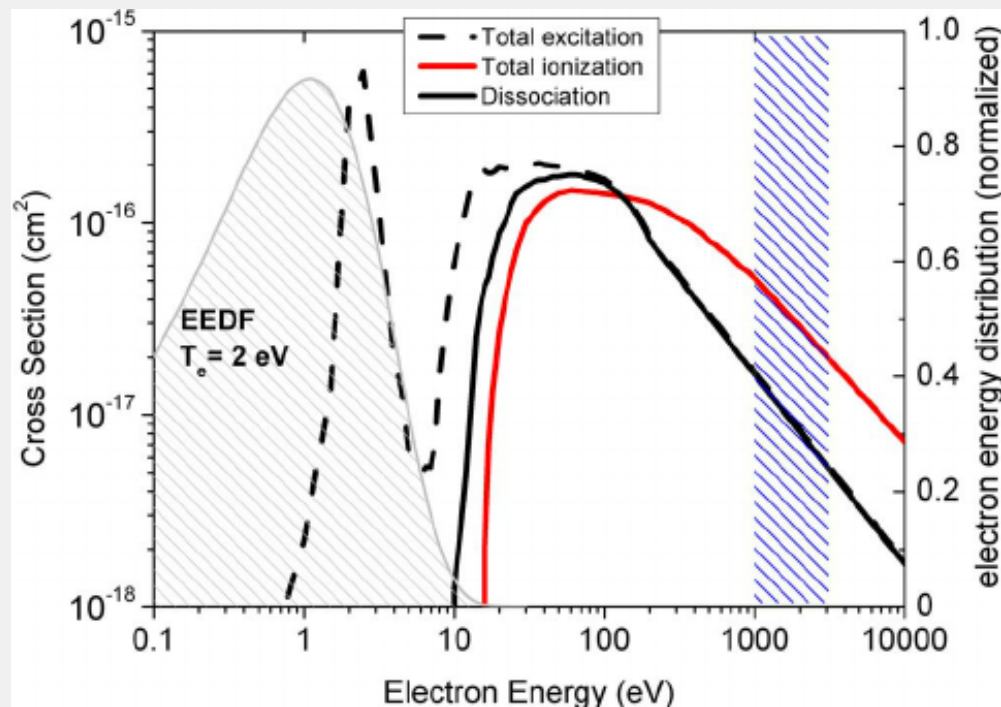
Atmospheric Heating

- ❖ Though less than 1.5% of TSI, wavelengths < 300 nm are primary source of atmospheric heating from 15–500 km.
- ❖ Excitation; dissociation; ionization
- ❖ Relaxation, association, recombination $\rightarrow$ heat

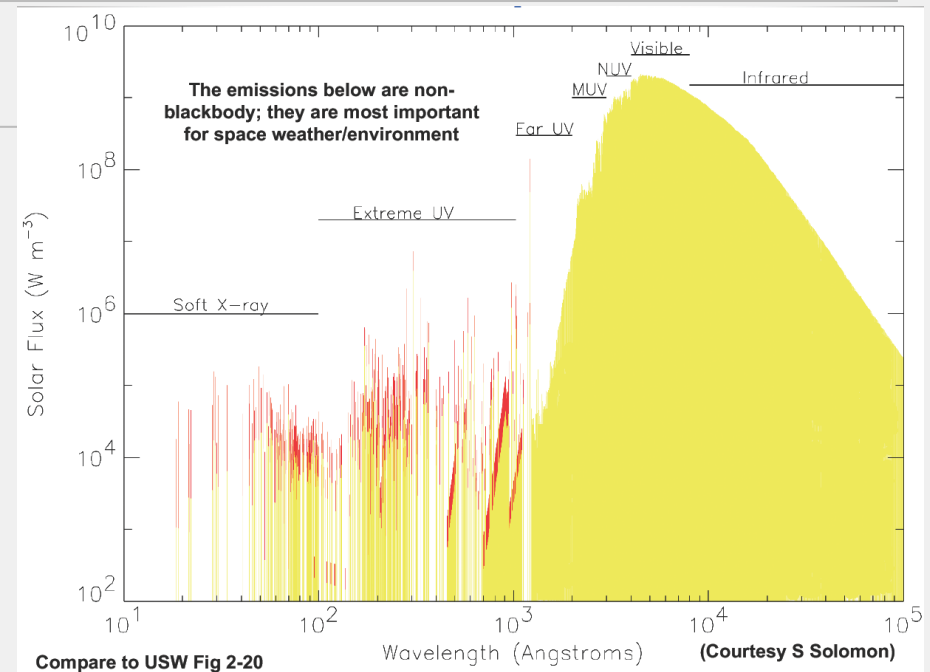


Ionization thresholds

- ❖ Require a minimum energy to free an electron from an atom or molecule
- ❖ Require a photon with at least this minimum energy: **"ionizing radiation"** or sometimes just **"radiation"**
- ❖ **Ionization cross section** provides energy-dependent picture of ionization probability



Electron Energy Distribution Function

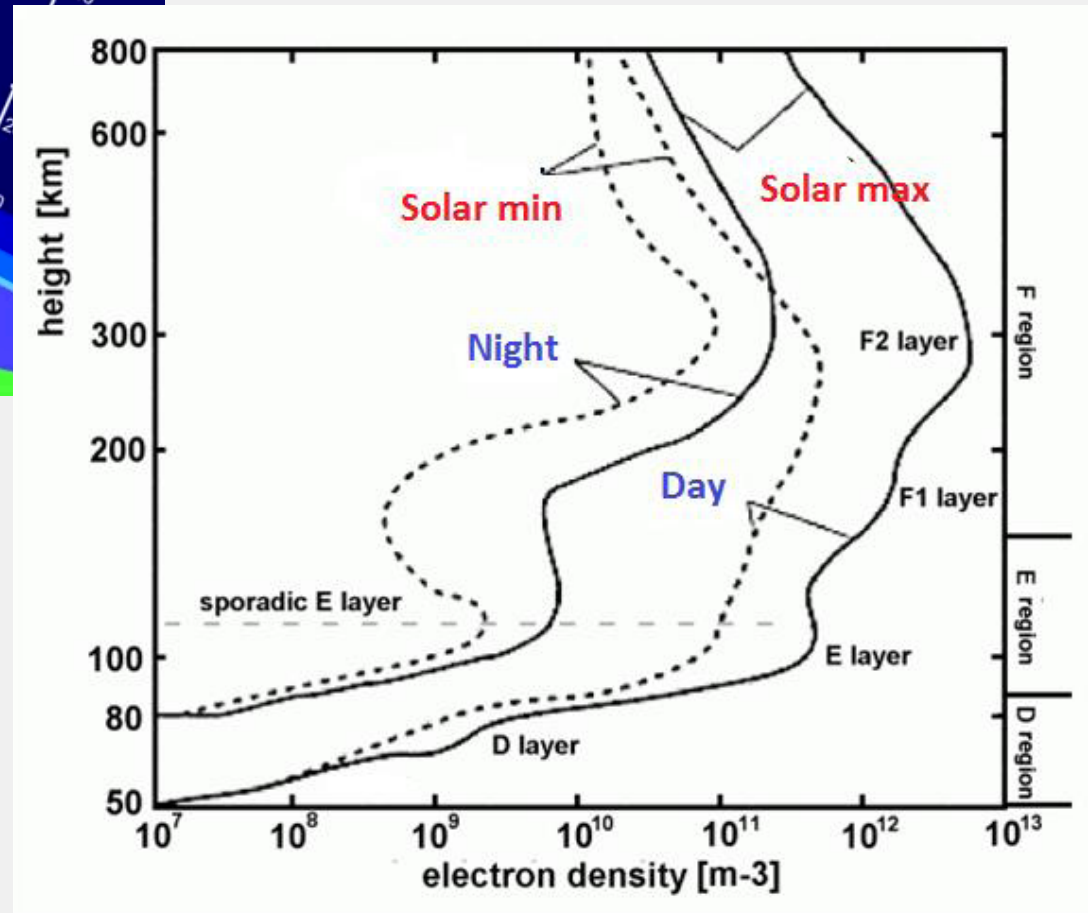
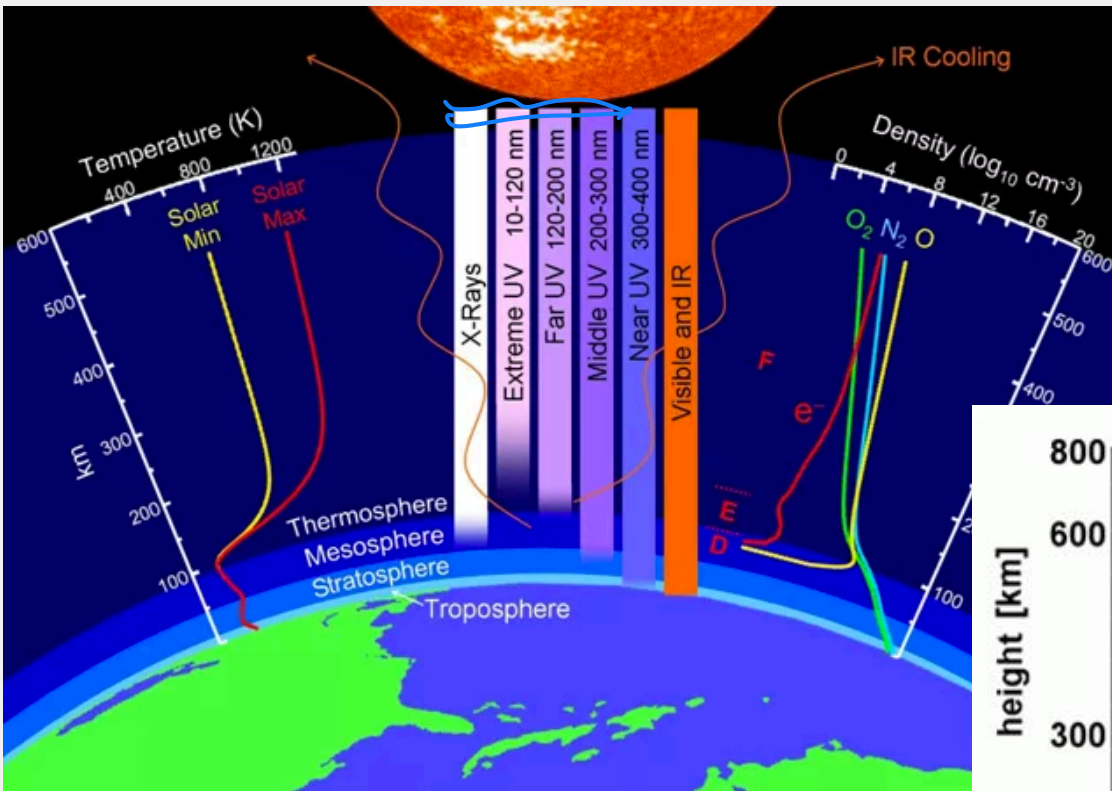


	Ionization Energy	Minimum photon wavelength
H	13.6 eV	91 nm (910 Å)
He	24.6 eV	50 nm
O	13.62 eV	91 nm
Ar	15.76 eV	79 nm
N ₂	15.6 eV	80 nm
O ₂	12.1 eV	103 nm

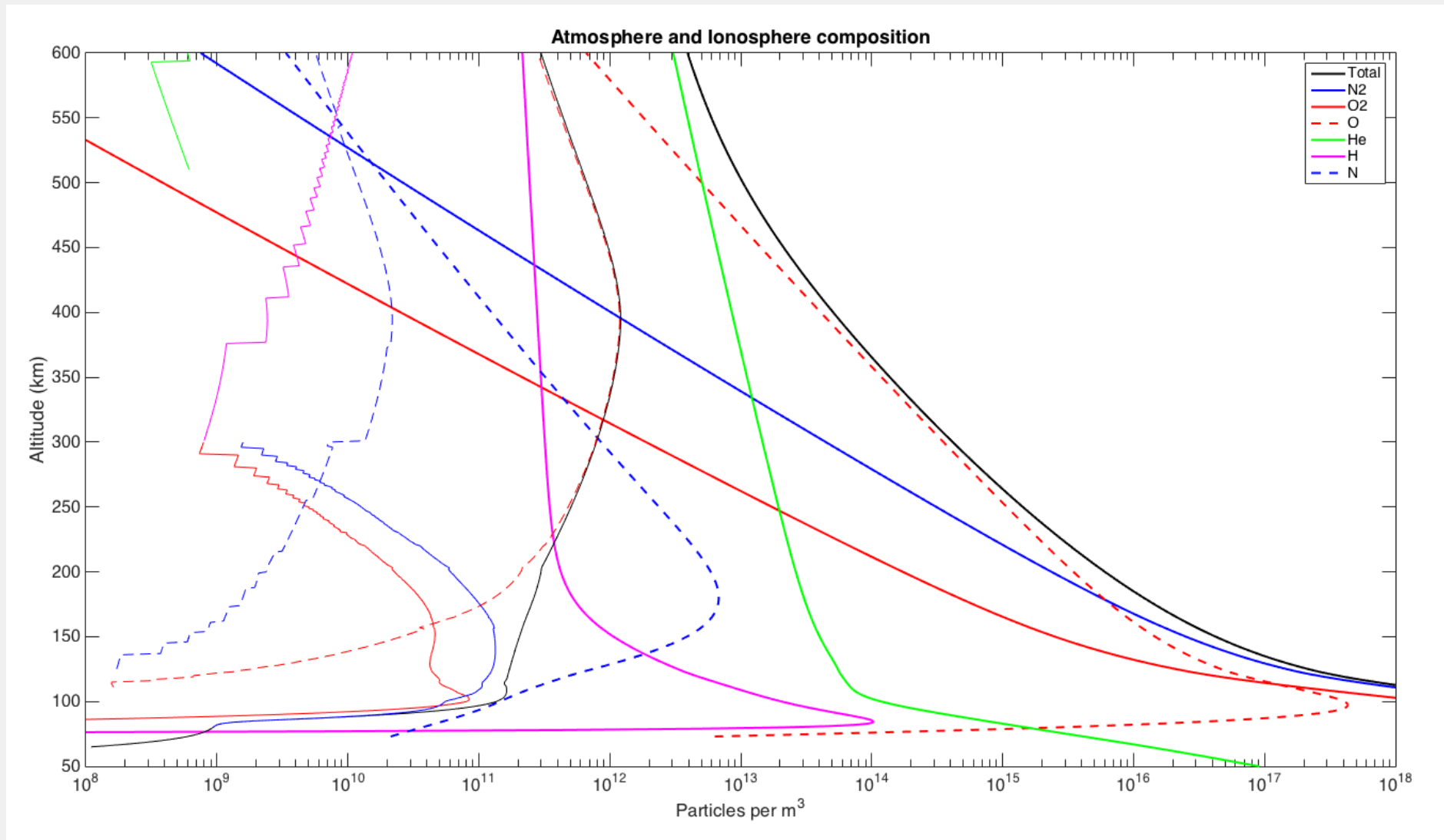
Do we concern about the solar flux in visible and infrared range? why?

Earth's Ionosphere

- ❖ Ionosphere altitudes and layers have a lot to do with where solar radiation is absorbed!

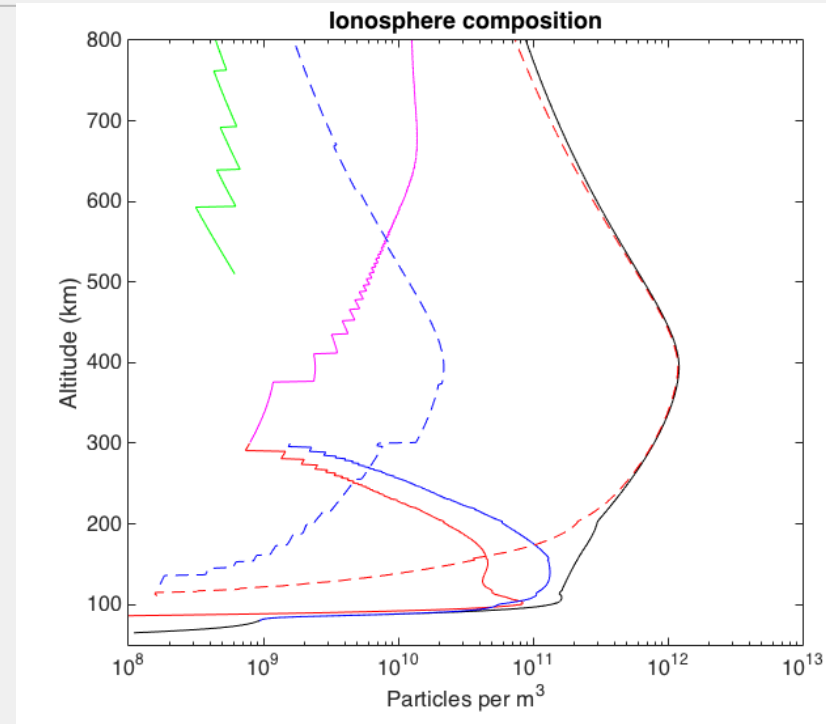


Ionosphere Composition and Density



Why is the ionosphere finite in altitude?

- ❖ The atmosphere is exponentially increasing all the way to the ground. What about the ionosphere?



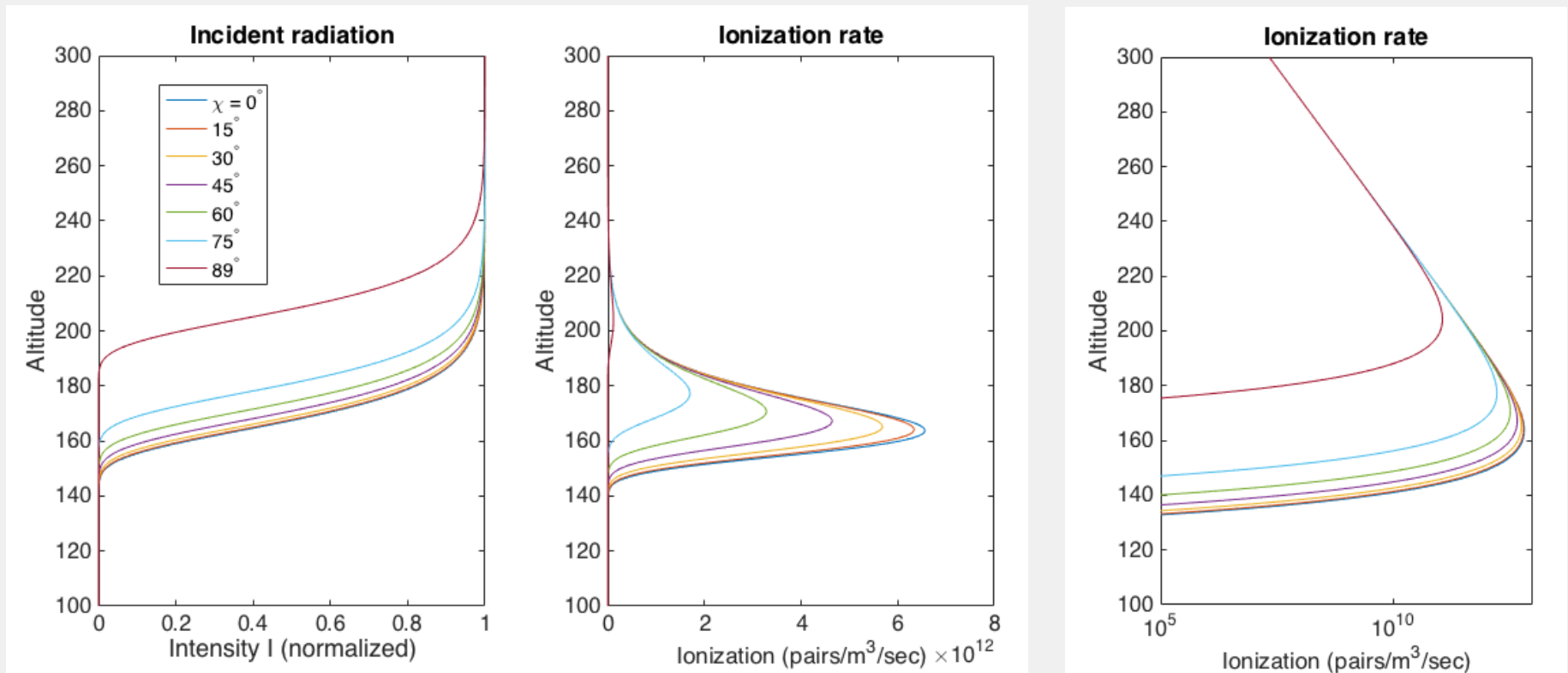
The relevant solar flux (in UV or shorter wavelengths) will become zero somewhere above the ground, why?

Chapman Layer

where, in altitude, will be the max production rate? How to figure this out?

Chapman Layer

- ❖ Production higher for lower zenith angle (i.e. noon)
- ❖ Peak in production is near where intensity is about **half** the incident value

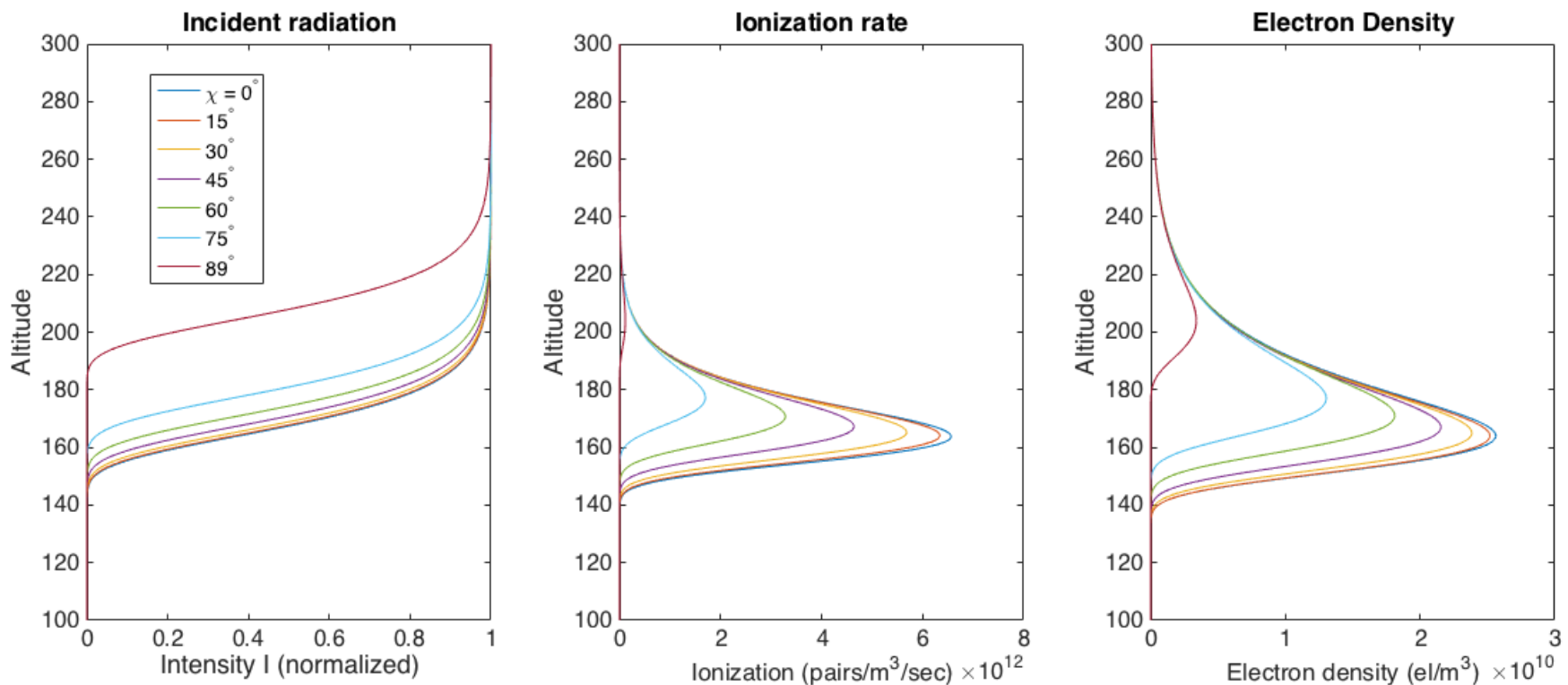


Ionization Chemistry

- ❖ Ionosphere is in equilibrium when ionization production and loss mechanisms balance
 - ❖ Production: **photoionization**; energetic particle precipitation; collisions
 - ❖ Loss: **recombination**; charge exchange; chemistry

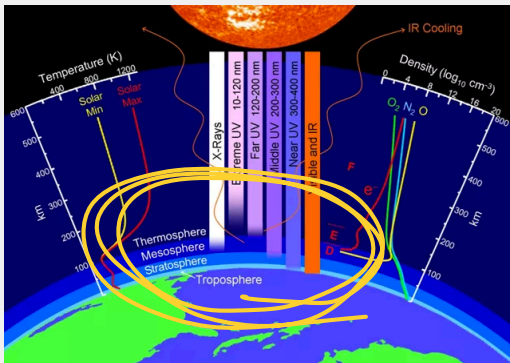
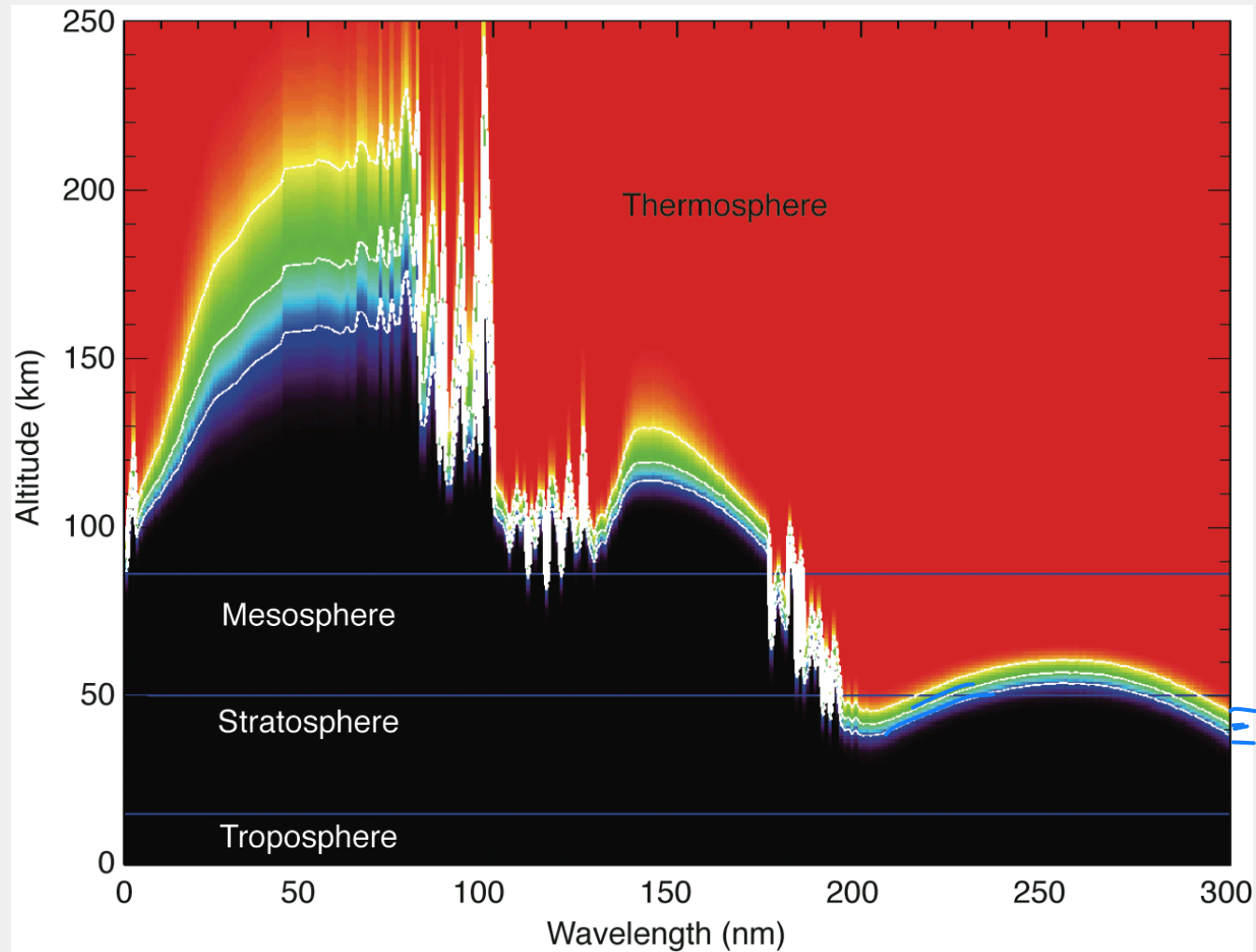
Electron Density Profile

- ❖ Balancing production (ionization) with loss (recombination), we get an equilibrium electron (or ion) density below
- ❖ Higher, less dense for increasing zenith angle
- ❖ Reminder: this is for a single species, and single photon wavelength!



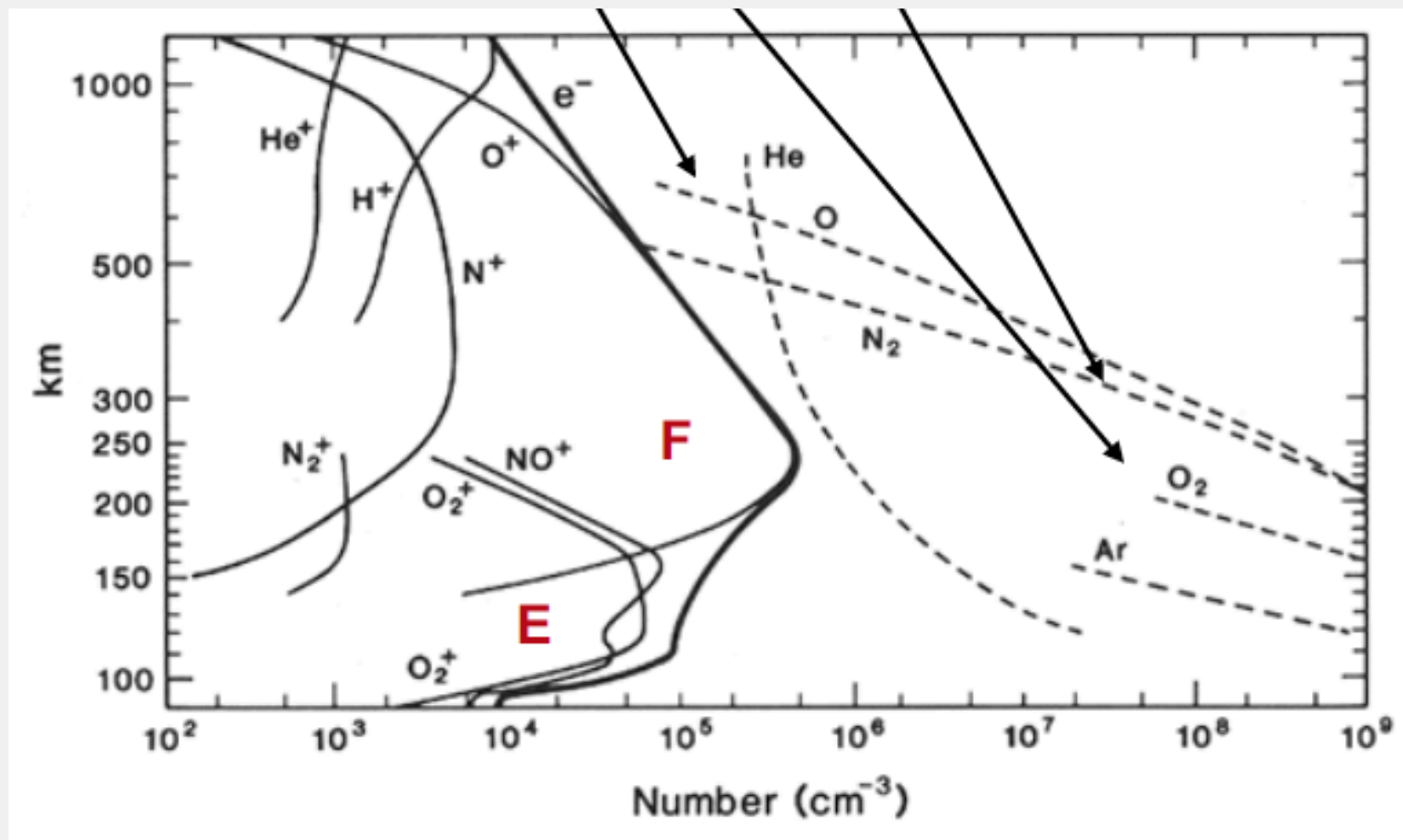
Ionosphere Layers

- ❖ Different wavelengths are absorbed at different altitudes, by different species
- ❖ $I(z)$ depends on wavelength-dependent absorption for each species
- ❖ $P(z)$ depends on wavelength-dependent ionization cross sections for each species
- ❖ Right: top white curve is $I(z)$ decay by $e^{-0.5}$; middle white curve by e^{-1} ; bottom white curve by $e^{-1.5}$
- ❖ Red areas: $I(z)$ is basically I_∞
Black areas: $I(z)$ is basically zero

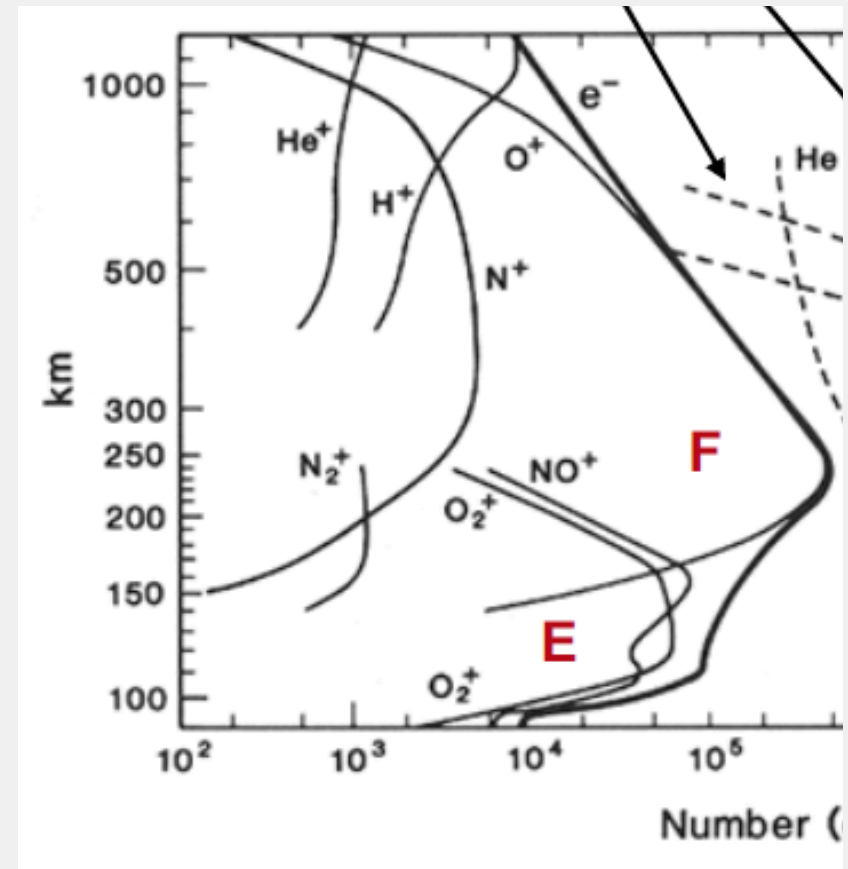


Another way of looking at it

- ❖ Ionosphere layers depend on composition (atmosphere) and radiation (solar spectrum and flux)
- ❖ D, E, and F regions are dominated by different ions, depending on the neutral species

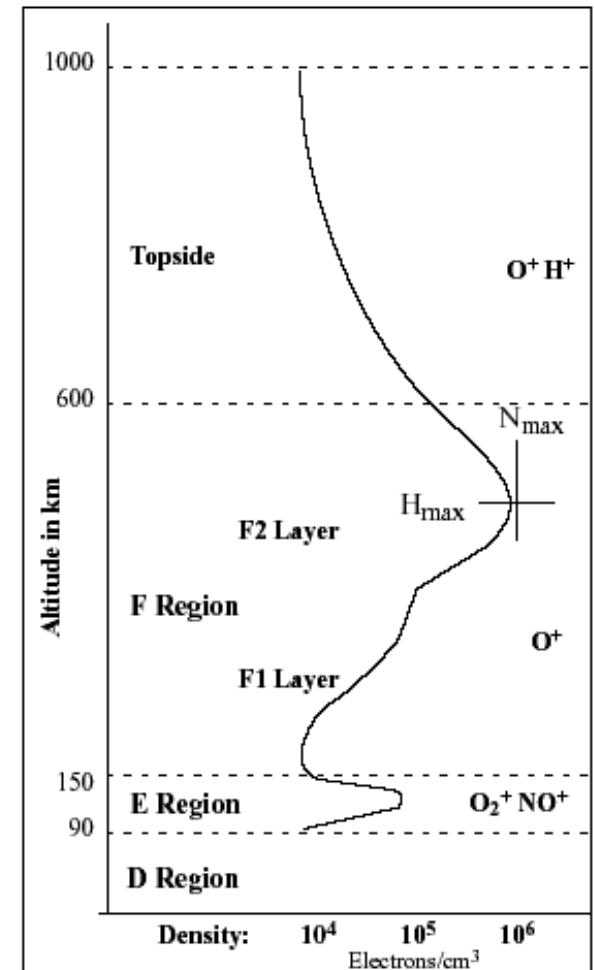


Primary Production / Loss channels

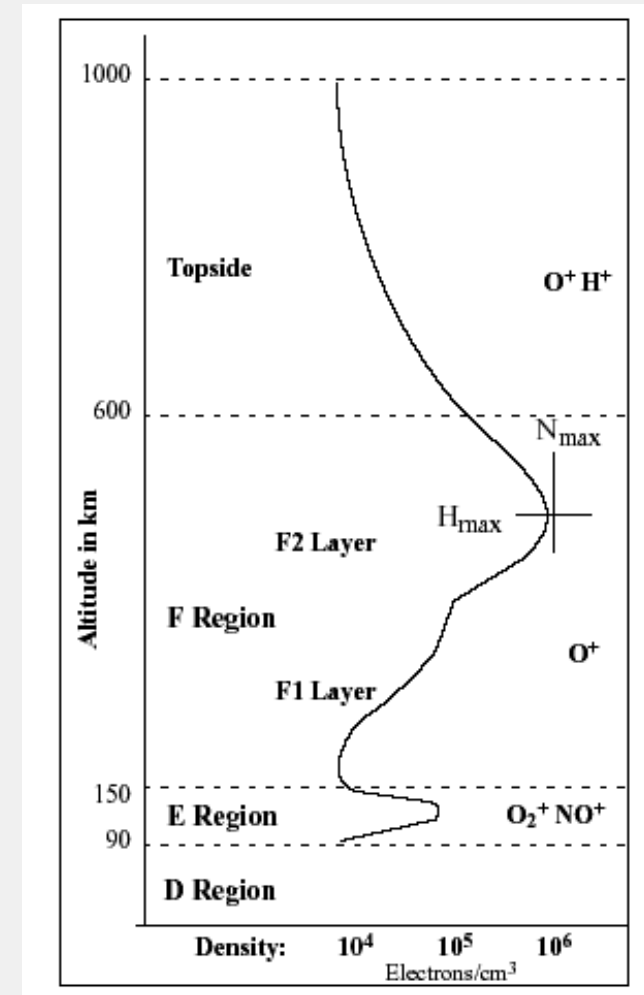


E-region

F-region

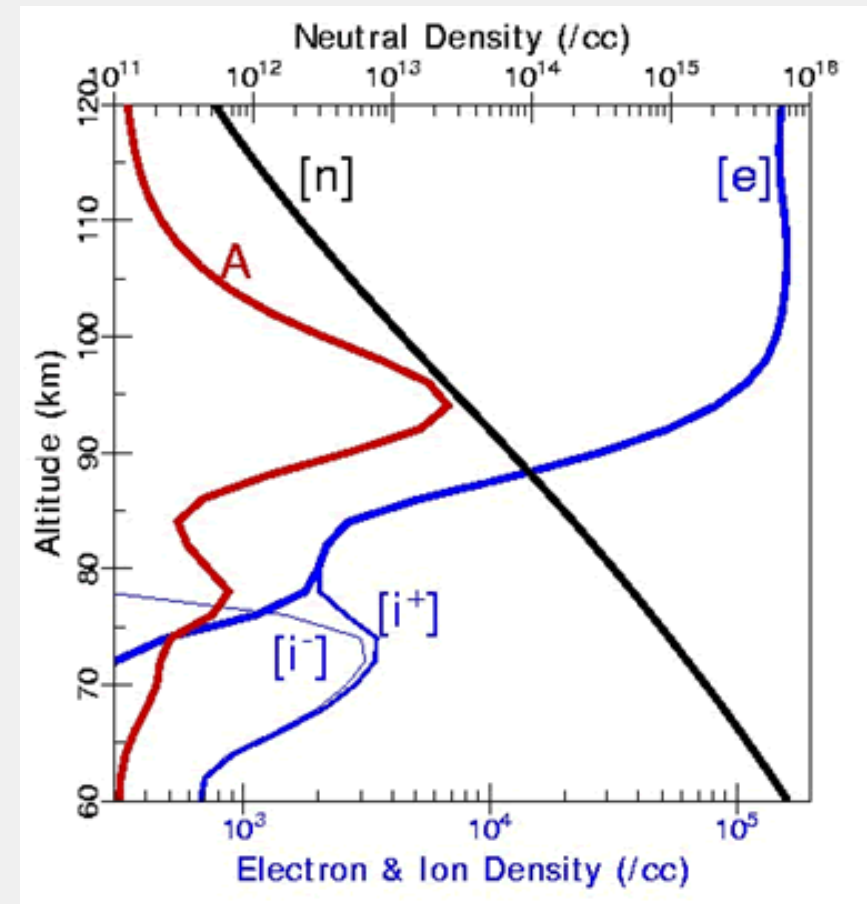


F-region



D-region

- ❖ D-region is known for low electron densities, light and heavy positive and negative ions, and complex chemistry. The neutral density is much higher there!
- ❖ Production:
 - ❖ $\text{N}_2 + \gamma \rightarrow \text{N}_2^+ + e^-$
 - ❖ $\text{O}_2 + \gamma \rightarrow \text{O}_2^+ + e^-$
 - ❖ $\text{NO} + \gamma \rightarrow \text{NO}^+ + e^-$
- ❖ Negative ions formed by attachment processes:
 - ❖ $\text{O}_2 + e^- + \text{O}_2 \rightarrow \text{O}_2^- + \text{O}_2$
 - ❖ $\text{O}_2 + e^- \rightarrow \text{O}_2^- + \gamma$
- ❖ Detachment:
 - ❖ $\text{O}_2^- + \gamma \rightarrow \text{O}_2 + e^-$
 - ❖ $\text{O}_2^- + \text{O}_2 \rightarrow \text{O}_2 + e^- + \text{O}_2$

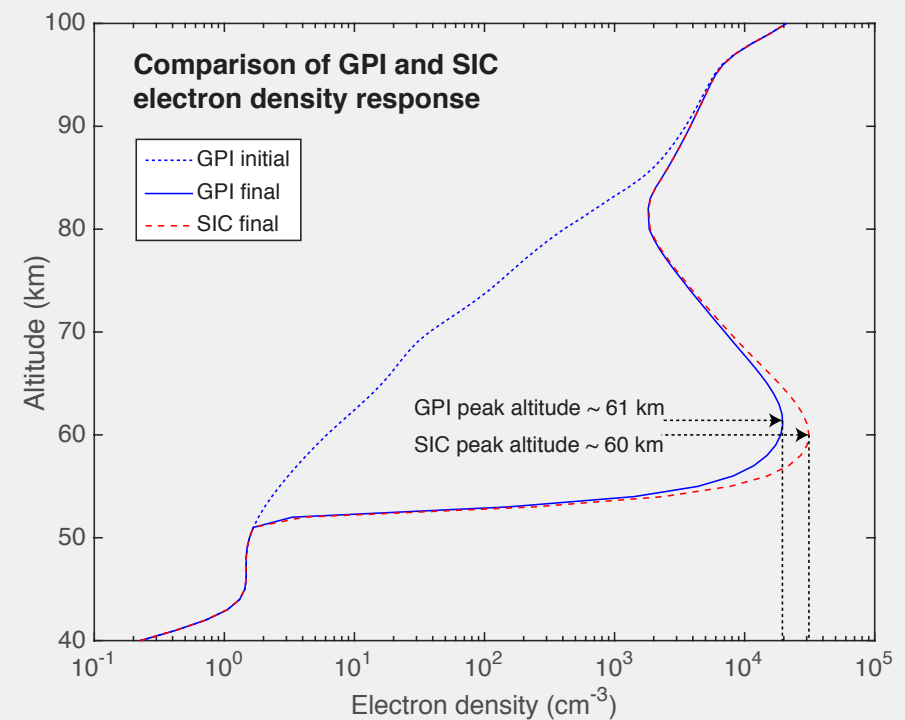


D-region chemistry and "cluster ions"

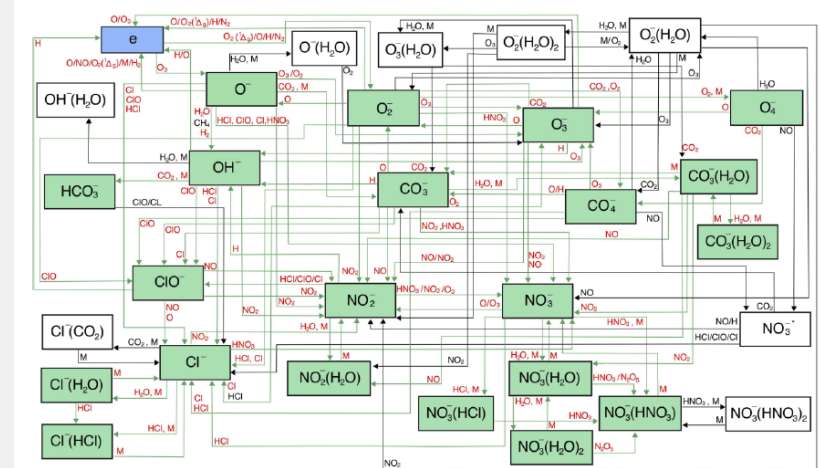
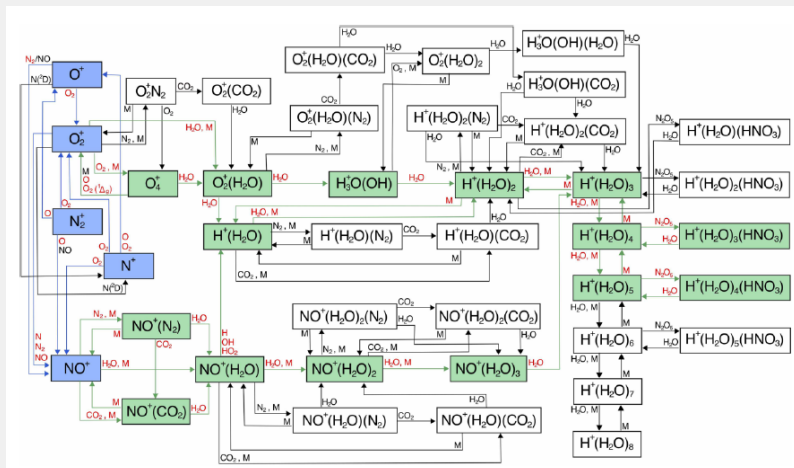
- ❖ D-region also contains heavy water cluster ions of the form $(\text{H}_2\text{O})_n\text{H}^+$
- ❖ Requires more complex chemistry models to evaluate n_e profiles

Glukhov Pasko Inan (GPI) 2009 model:

$$\begin{aligned}\frac{dN_e}{dt} &= Q - \beta N_e + \gamma N^- + \gamma_x N_x^- - (\alpha_d N^+ + \alpha_d^c N_x^+) N_e \\ \frac{dN^-}{dt} &= \beta N_e - \gamma N^- - \alpha_i (N^+ + N_x^+) N^- - A N^- \\ \frac{dN_x^-}{dt} &= -\gamma_x N_x^- - \alpha_i (N^+ + N_x^+) N_x^- + A N^- \\ \frac{dN^+}{dt} &= Q - \alpha_d N_e N^+ - \alpha_i (N^- + N_x^-) N^+ - B N^+ \\ \frac{dN_x^+}{dt} &= \alpha_d^c N_e N_x^+ - \alpha_i (N^- + N_x^-) N_x^+ + B N^+\end{aligned}$$

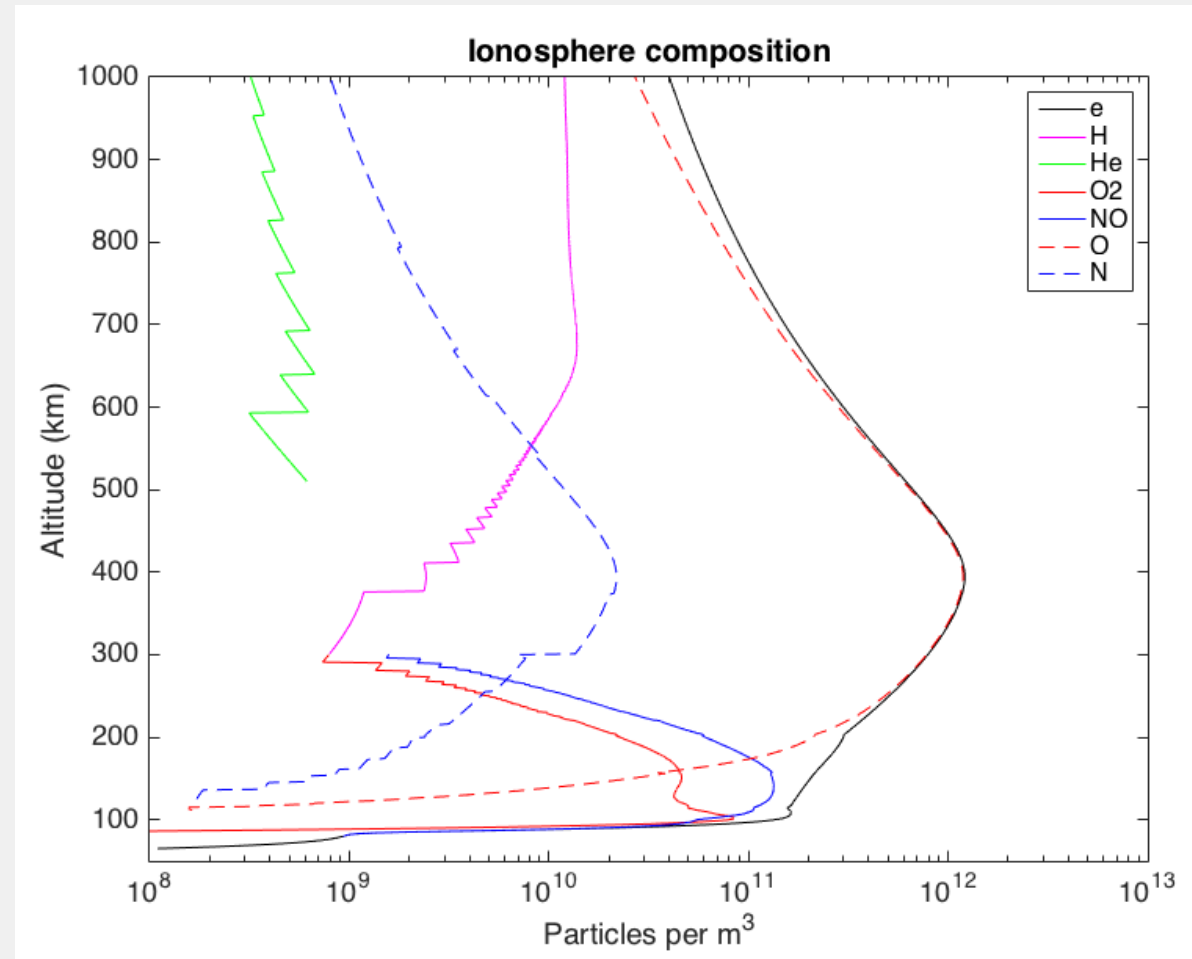
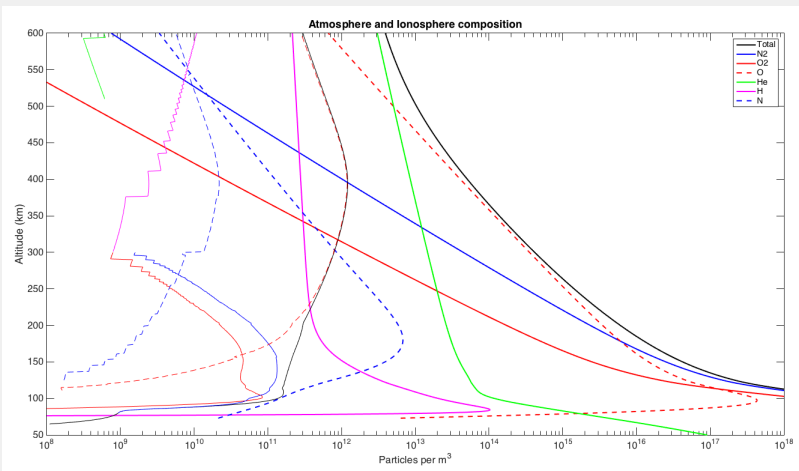


Sodankylä Ion and Neutral Chemistry (SIC) model:



Topside Ionosphere

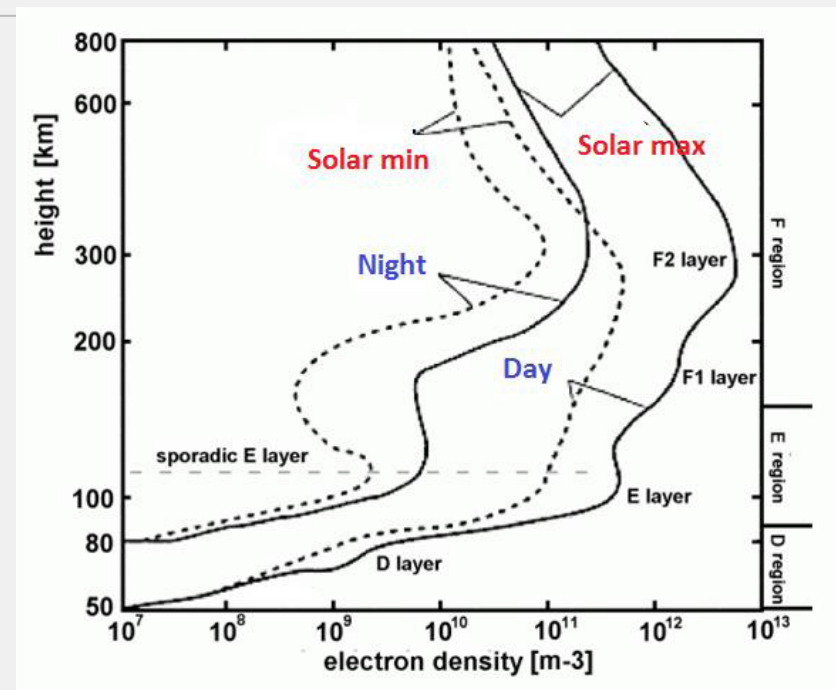
- ❖ Above ~350 km, densities are so low that ions are not dominated by chemistry, but diffusive equilibrium



analogous to homosphere / heterosphere: bottom has turbulent mixing, top has diffusive equilibrium. In ionosphere case, lower altitudes have chemical equilibrium, higher has diffusive equilibrium.

Summary of Layers

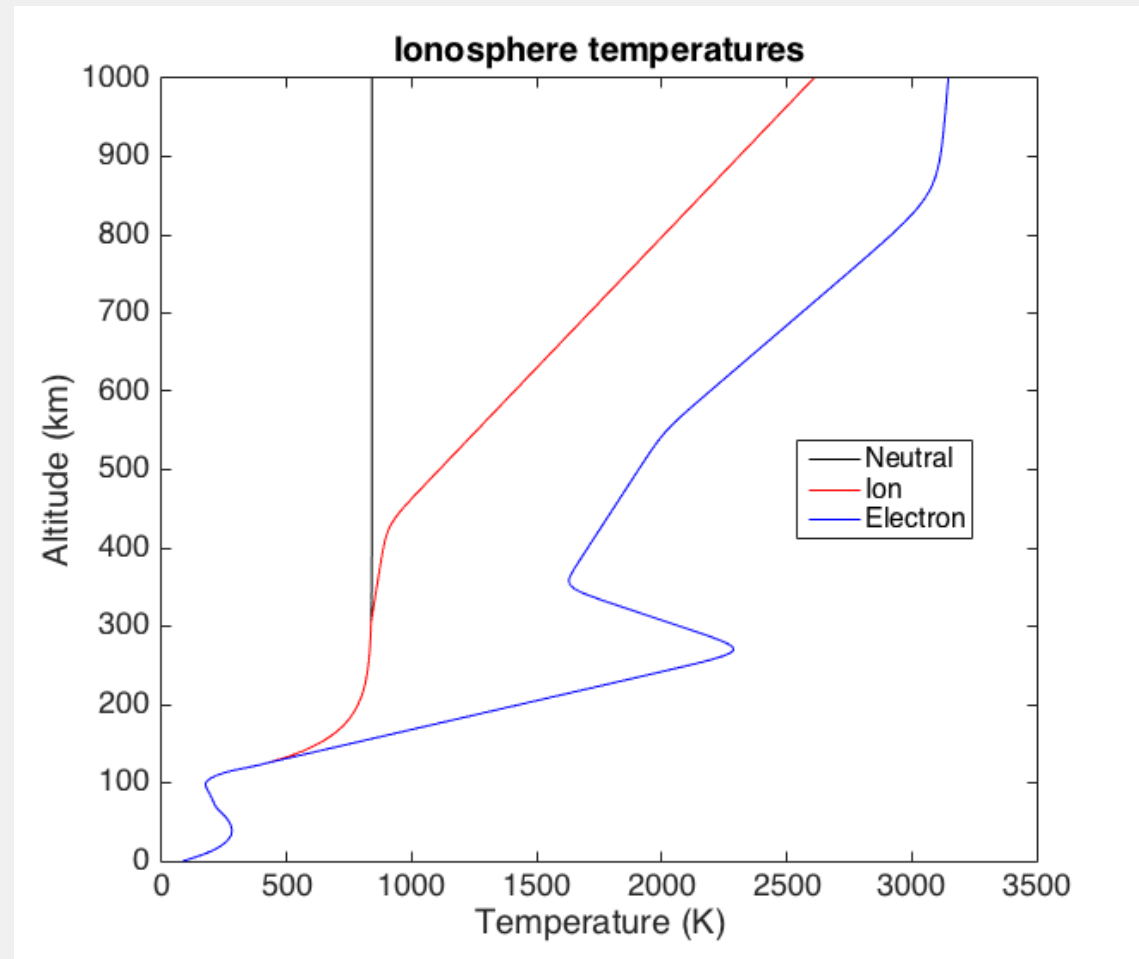
- ❖ Layers are dominated by different ions
- ❖ Ion densities controlled by balance between production and loss; depends on chemical reactions in each altitude range
- ❖ Decay of layers at night depends on recombination rates and densities: low densities at high altitudes mean few collisions



Ionospheric Layer	Altitude Range (km)	Major Constituents	Notable Characteristics
D	70–90	NO ⁺ O ₂ ⁺ (molecular)	Disappears (recombines) very rapidly—minutes after sundown
E	90–140	O ₂ ⁺ (molecular) NO ⁺	Recombines rapidly—often disappears before midnight
F1	140–200	O ⁺ (atomic) NO ⁺	Mostly recombines after sundown, but pockets of ionization may remain
F2	200–400	O ⁺ (atomic)	Persistent because of low collision rates, but density decreases after sundown
Topside	> 400	O ⁺ (atomic) H ⁺	Merges into the plasmasphere, atomic oxygen dominates at lower altitudes, and hydrogen dominates at higher altitudes.

Ionosphere Temperatures

- ❖ Below 110 km, temperatures are made equal by collisions
- ❖ Above ~110 km, collisions are rare, so each species gets its temperature through different heating processes (radiation absorption, convection, etc)
- ❖ Above ~110 km: ions, electrons, and neutrals have different temperatures
 - ❖ Remember: this simply means they have different velocity / energy distributions





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Ionosphere Variations

Transport

- ❖ Variety of processes move plasma in the ionosphere:
 - ❖ **Winds:** neutral winds drag ions along with them, if collision frequency is high enough
 - ❖ **Drifts:** various forces cause plasma to “drift”: electric and magnetic fields; gravity; pressure; etc.
- ❖ **These contribute to complex spatial and temporal variations in the ionosphere**